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Elevated levels of anthropogenic antibiotic resistance gene marker, *sul1*, linked with extreme fecal contamination and poor water quality in wastewater-receiving ponds

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Abstract

In several low- and middle-income countries, such as India, the rapid construction of toilets to combat open defecation has not been matched with adequate wastewater treatment, resulting in extreme fecal contamination of the receiving environments. The sewage-receiving surface water bodies, typically close to the residences, are a potential hotspot for disease transmission and antibiotic resistance. Water, soil, and sediment samples from seven wastewater-receiving ponds (WRPs) were analyzed for water quality, chlorophyll-*a*, fecal contamination (*yccT* for *Escherichia coli*), 16S rRNA gene copies, and anthropogenic antibiotic resistance gene markers—*sul1* and *intI1*. These WRPs were contrasted with two ponds that did not directly receive sewage. The water quality in the WRPs was comparable to raw sewage (BOD: 210–380 mg/L; COD: 350–630 mg/L; total-N: 100–190 mg/L; and total-P: 6–21 mg/L), and the relative levels of the DNA marker of *E. coli* were very high (*yccT*: 0.1% to ~100% of total bacterial count) indicating extreme fecal contamination. The relative levels of *sul1* and *intI1* were 1–3 orders of magnitude higher in WRPs (*sul1*: 0.32%–10% of total bacterial count; and *intI1*: 0.2%–5% of total bacterial count) compared to the ponds that did not receive sewage directly. The relative levels of *sul1* correlated with the DNA marker for the fecal indicator, *E. coli* (p -value < 0.05; r = 0.50; Spearman's rank correlation), and poor water quality.

1 | INTRODUCTION

Increased access to toilets in some low- and middle-income countries (LMICs) (Cronk et al., 2021; Exum et al., 2020; O'Reilly et al., 2017) has not been accompanied by adequate wastewater collection and treatment systems (Rashid & Pandit, 2019). Consequently, the effluents from the toilets often end up in small pools, drains, and low-lying areas, including local ponds that traditionally assisted with water harvesting and conservation, especially in rural, semi-, and peri-urban areas (Ghosh et al., 1999). These wastewater-receiving ponds

(bodies) (WRPs) are frequently near residences and attract pathogen transmission vectors. Although the water in these wastewater-receiving water bodies is not used for direct consumption, it is susceptible to pathogen transmission via vectors, such as rats and flies (Sclar et al., 2016). The related chronic water-borne infections are amongst the leading drivers of public health burden due to infectious diseases in many LMICs (Goddard et al., 2020) and are strongly associated with increased childhood wasting and stunting (Arankalle et al., 1994; Iips, 2017; Khan et al., 2018; Swaminathan et al., 2019). Consequently, any improvement in

primary healthcare and sanitation (Ercumen et al., 2018) can be overturned by accidental or indirect exposure to heavily contaminated WRPs (Goddard et al., 2020; Luby et al., 2018; Null et al., 2018).

The widespread fecal contamination of water bodies with untreated or partially treated sewage (Binh et al., 2018) co-occurs with heavy misuse of antibiotics in India (Barker et al., 2017), which are eventually discharged into the WRPs (Khare et al., 2019). Both the environmental contamination with antibiotics and their derivatives and fecal contamination (Bengtsson-Palme et al., 2014) are strongly linked with the selection and emergence of antibiotic resistance (Karkman et al., 2019). As domestic wastewater carries commensal gut bacteria and pathogens, along with co-selectors of antibiotic resistance: antibiotics, heavy metals, and certain personal care products (Gorham et al., 2017; Wu et al., 2022), these WRPs are not only potential reservoirs of pathogens but also possibly a breeding ground for antibiotic resistance (Arankalle et al., 1994; Schaible & Kaufmann, 2007). Such wastewater-receiving water bodies can also contaminate other water resources (McGuinness et al., 2020), including shallow groundwater, a significant drinking water source in India, and other LMICs (Sorensen et al., 2016). In the context of antibiotic resistance, much emphasis has been laid on the major water bodies, such as rivers, whereas the informal WRPs are yet to be investigated despite their prevalence across LMICs and their proximity to residences, where toilets are rapidly being built (Abubakar, 2018) but without the accompanying wastewater treatment infrastructure. Although the rapid construction of toilets is well intended (Abubakar, 2018), the absence of adequate wastewater treatment and management facilities could defeat the purpose and even reverse the benefits of increased toilet availability and use (Jangra et al., 2016).

It is important to note that, although both the WRPs and wastewater stabilization ponds receive domestic wastewater, they differ from each other as the former is not engineered for wastewater treatment and has infinite hydraulic retention time (Abis & Mara, 2005; Vymazal, 2007). The WRPs is a convenient, albeit transitory, and ineffective solution for managing domestic wastewater, but it is unknown whether the WRPs can perform comparably to wastewater stabilization ponds (Abis & Mara, 2005; Vymazal, 2007). To the best of our knowledge, there has not been any scientific investigation into the water quality of the WRPs and the persistence of fecal bacteria in these water bodies.

In this study, we investigated water quality, antibiotic resistance potential, and fecal markers in WRPs that were neither designed for sewage treatment nor ever cleaned up. We sampled seven WRPs and contrasted them with two ponds (sites H and I) that did not directly receive domestic wastewater and were used for irrigation (P-I) and pisciculture (P-P), respectively. We observed that the water from the targeted WRPs

Core Ideas

- Discharge of untreated sewage in local ponds is linked with 3 log₁₀ higher levels of antibiotic resistance potential.
- There is no evidence of any fecal degradation in wastewater-receiving ponds.
- Water quality in wastewater-receiving ponds is at par with medium-to-high strength raw sewage.

was typically not used for drinking or fishing and was only occasionally used for irrigation (Datta et al., 2021; Maiga et al., 2015). In the water, sediment, and adjoining soil samples, we quantified the DNA markers of the fecal indicator (*Escherichia coli*) to investigate the persistence of fecal bacteria; 16S rRNA gene copies to track the trends in the total bacterial count; and the anthropogenic antibiotic resistance gene markers—*int11*, *sul1*, to investigate the overall trends in antibiotic resistance potential in WRPs. Furthermore, we analyzed the physicochemical parameters of the sampled water: alkalinity, COD, BOD, chlorophyll-*a*, ammoniacal nitrogen, nitrate, phosphate, and DO. The overarching objective was to estimate the potential for antibiotic resistance, fecal persistence, and water quality deterioration in surface water bodies that are receiving wastewater. The extent of contamination in WRPs was worse than in other sewage-contaminated environments reported in the literature (Osmanov et al., 2020), with the water quality comparable to high-strength sewage (Hahn & Figueroa, 2015).

2 | MATERIALS AND METHODS

2.1 | Site description and sampling

2.1.1 Site description

Nine ponds from nine villages in the Haridwar district, located on a stretch of 15.7 km along the highway (NH 58) in Uttarakhand, were selected for this study (Figure S1 and Table S1). All the sampled sites were located on the north Indian plain near the NH-58 highway and the Upper Ganga Canal. Seven of these nine ponds received domestic wastewater and are referred to as WRPs. Briefly, 30–50 households discharged their domestic wastewater in the seven WRPs, including any wastewater from the livestock that was kept within the premises, which was also discharged through underground sewage pipes into the WRPs (sites A–G). Besides the domestic wastewater (from both the humans and the livestock inside the premises), the WRPs received water only from surface

run-off post precipitation events, except for the WRP at site G, which also received wastewater from a dairy processing unit. As six of the WRPs (sites A–F) received wastewater of comparable origins, they serve as independent replicates. The households discharged the wastewater via private lines that were underground and inaccessible for sampling. All the WRPs had visibly abundant algal growth (Figure S2 and Table S1), whereas this was not the case for the two ponds that did not receive wastewater (Figure S2 and Table S1). The residents also discarded domestic solid waste in the vicinity of the WRP. The households were typically located 3–5 m from the WRP (Figure S2).

Sites H and I did not directly receive domestic wastewater and are hereafter referred to simply as ‘control ponds’: Water from site H was used to irrigate a nearby farm (P–I). Pond at the site I was employed for pisciculture (P–P) (Figure S2). During the sampling, we observed that P–P (site I) was also used for swimming by local children. In the target region, all ponds except the two control ponds had either dried up or been converted into WRPs. Possible vectors of disease transmission—rats and flies—were present at all the WRP sites. Detailed information on the sampling sites is available in Section SI.1.

2.1.2 Sampling

The water, sediment, and adjoining soil samples from the seven WRPs and two ponds were collected on a cool (17–19°C) sunny day. Water samples were collected in sterile 50 mL conical sterile polypropylene centrifuge tubes (Abdos, Bhagwanpur, India) in triplicates for analyses of physicochemical parameters and genetic markers. Local soil and sediment samples were collected using an ethanol sterilized spatula in sterile 50 mL conical sterile polypropylene centrifuge tubes (Abdos, Bhagwanpur, India) in triplicates from all the ponds. A volume of 50-mL LC–MS grade water was used as a field blank at all sampling sites. The samples were collected and transported to the laboratory on ice within a day. The analysis of the physicochemical parameters commenced within 24 h of sampling. For molecular-biological analysis, the water samples were filtered aseptically through sterile 0.22 µm mixed cellulose–ester filter papers (Axiva Siche, Sonipat, India) on the day of sampling and stored at –20°C until DNA extraction.

2.2 | Analysis of water quality

For analysis of physiochemical parameters, 50 mL of water was used to analyze alkalinity, COD, BOD, chlorophyll-*a*, NH₄–N, NO₃–N, PO₄–P, and DO. The DO was measured using a digital probe meter (HI media Labs, Mumbai, India).

On the day of sampling, COD, NH₄–N, NO₃–N, and PO₄–P were measured in the lab with the help of a UV spectrophotometer by a closed reflux method (standard method 5220 D) (Matthews, 2014), phenate method (standard method 4500-NH₃) (APHA method 4500; APHA et al., 2005), screening method (standard method 4500-NO₃-B) (APHA method 4500; APHA et al., 2005), and stannous chloride method (standard method 4500 P) (Baird et al., 2012), respectively. Alkalinity was measured with the help of the titration method following the APHA standard method 2320 B (Baird et al., 2012). Chlorophyll-*a* was measured with a UV spectrophotometer using the acetone method (Ji et al., 2018). BOD₅ was determined using the standard method 5210 B (Gooch, 2011).

2.3 | Molecular-biological analysis

For DNA extraction from the water samples, the filter papers were aseptically torn using flame-sterilized forceps and then used as samples with the DNeasy PowerSoil Kit for soil (Qiagen, Hilden, Germany). The sediment and local soil samples were directly used as samples for DNA extraction with the DNeasy PowerSoil Kit for soil (Qiagen, Hilden, Germany). The DNA was extracted according to the manufacturer's protocol (Qiagen, Hilden, Germany) with a modification at the cell-lysis step: Tissue Lyser (Qiagen, Hilden, Germany) was used for cell lysis at 25 Hz for 30 s. The targeted genetic markers were quantified using quantitative polymerase chain reaction (qPCR) on Rotor-Gene Q (Qiagen, Hilden, Germany). The standards for the qPCR were prepared by cloning the target amplicon into chemically competent TOP10 cells using a TOPO TA Cloning Kit (Invitrogen, CA, USA) as per the manufacturer's instructions (Table S2). After testing the specificity of the inserts as described elsewhere (Bhaskar & Singh, 2021), the M13 PCR amplicon of the clone was quantified using NanoDrop (Thermo Fisher Scientific, MA, USA) and diluted serially to prepare qPCR standards (10⁸–10² genes copies per µL) that were included in each run. The extracted DNA was diluted to reduce PCR inhibition, which was determined by generating a dilution curve using qPCR targeting the 16S rRNA gene (1:50; 1:200 for water samples, 1:100 for soil, and 1:100 for sediment). Each qPCR run included a negative control, where molecular biology-grade water was used instead of DNA. If the *Ct* value of the negative control was less than the limit of detection, the qPCR assay was rejected and repeated (Table S3).

2.4 | Statistical analysis

Origin (Pro) 2018 (Origin Lab Corporation, Northampton, MA, USA) was used for data visualization. The gene data

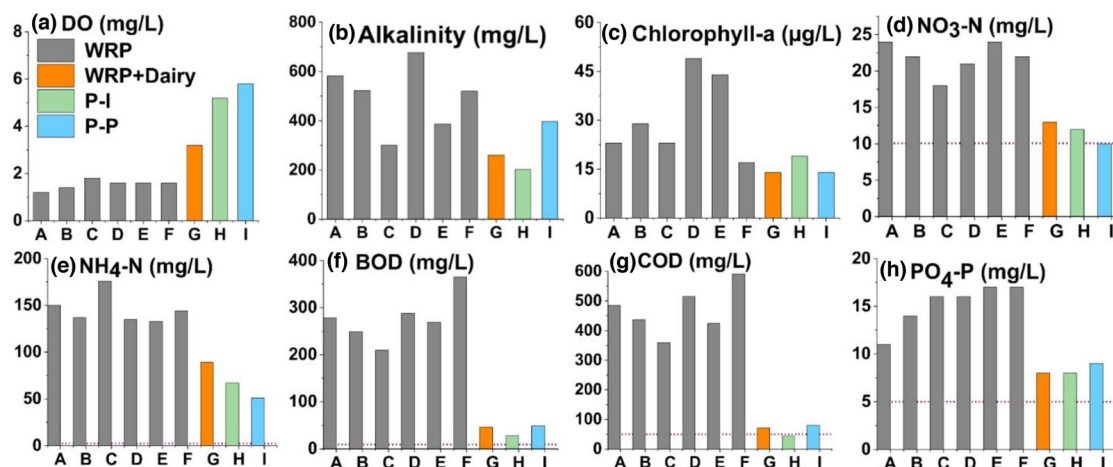


FIGURE 1 Water quality parameters: water samples from seven wastewater-receiving ponds (WRPs) and two control ponds were analyzed for (a) DO, (b) alkalinity, (c) chlorophyll-*a*, (d) $\text{NO}_3\text{-N}$, (e) $\text{NH}_4\text{-N}$, (f) BOD, (g) COD, and (h) $\text{PO}_4\text{-P}$. WRP + dairy (WRP + dairy wastewater) represents site G, which received wastewater from a dairy processing unit and domestic wastewater. P-I represents the control pond whose water was used for irrigation, whereas P-P represents the control pond where pisciculture was practiced. The dashed orange line represents the Indian standards for treated effluent (CPCB, 2021).

were transformed to log10-scale before analysis. The copy number of the genetic marker in any sample detected by qPCR was defined as the ‘absolute level’ of the relevant gene. The relative levels of a gene were defined as $\log_{10}(\text{absolute gene copies/L}) - \log_{10}(16\text{S rRNA gene copies/L})$, they were referred to as ‘relative levels’ of the relevant gene. As the qPCR data was not normally distributed, as determined by the Shapiro–Wilk test and q–q plots, nonparametric tests were used throughout. Spearman’s rank correlation among physicochemical parameters such as alkalinity, COD, BOD, chlorophyll, $\text{NH}_4\text{-N}$, $\text{NO}_3\text{-N}$, $\text{PO}_4\text{-P}$, and DO; and targeted genes (16S rRNA gene, *sul1*, *int11*, and *yccT*) was used. The Mann–Whitney *U* test was performed to compare the levels of the 16S rRNA gene copies, *yccT*, *sul1*, and *int11*. The statistical analysis’s significance threshold was set at alpha value = 0.05. All statistical analyses were conducted using Microsoft Excel (2018).

3 | RESULTS

3.1 | Water quality of the ponds receiving untreated domestic wastewater

In WRPs, the BOD and COD levels were comparable to that of high-strength raw sewage (Guwy et al., 1999; Hahn & Figueroa, 2015; Yadav et al., 2018): BOD ranged from 210 to 372 mg/L (Table S4 and Figure 1F) and the COD ranged from 359 to 610 mg/L (Table S4 and Figure 1G). The $\text{PO}_4\text{-P}$ levels (7–20 mg/L) in both the WRPs and ponds also exceeded the standard for treated wastewater (5 mg/L) (Sawal, 1986) (Table S4 and Figure 1H). Sites H and I (ponds) had higher DO (3–6 mg/L) relative to all the WRPs ($\text{DO} < 2$ mg/L) (Table S4 and Figure 1A), whereas the total alkalinity was usually higher in

WRPs (Table S4 and Figure 1B). Chlorophyll-*a* levels ranged from 16 to 50 $\mu\text{g/L}$ in WRPs and 11 to 21 $\mu\text{g/L}$ in the ponds (Table S4 and Figure 1C), indicating that all the WRPs were eutrophic, whereas the ponds were mesotrophic (Sepulveda-Jauregui et al., 2013). The $\text{NO}_3\text{-N}$ levels (9–30 mg/L) in both WRPs and ponds exceeded the standard set for treated wastewater effluent at 10 mg/L (Sawal, 1986), except for one pond that was used for pisciculture (site I) (Table S4 and Figure 1D). The $\text{NH}_4\text{-N}$ levels (45–180 mg/L) were higher than the Indian standard set for treated wastewater (> 5 mg/L) (CPCB, 2021) in all the ponds (Table S4 and Figure 1E). The water in neither the ponds nor the WRPs met the latest CPCB guidelines for treated wastewater (CPCB, 2021).

The levels of chlorophyll-*a* correlated positively and significantly with the levels of 16S rRNA gene copies (p -value < 0.001 and $r = 0.63$), *yccT* (p -value < 0.01 ; $r = 0.54$), relative levels of *sul1* (p -value < 0.05 ; $r = 0.48$), alkalinity (p -value < 0.05 ; $r = 0.47$), $\text{NH}_4\text{-N}$ (p -value < 0.05 and $r = 0.48$), $\text{NO}_3\text{-N}$ (p -value < 0.001 and $r = 0.64$), and $\text{PO}_4\text{-P}$ (p -value < 0.01 and $r = 0.56$) and correlated negatively with DO (p -value < 0.05 ; $r = 0.50$) (Spearman’s rank correlation) (Figure 2, Table S5). The COD correlated positively with $\text{NH}_4\text{-N}$ (p -value < 0.001 and $r = 0.70$), $\text{NO}_3\text{-N}$ (p -value < 0.0001 and $r = 0.71$), $\text{PO}_4\text{-P}$ (p -value < 0.0001 and $r = 0.70$), and negatively with DO (p -value < 0.0001 and $r = 0.72$) (Figure 2, Table S5). BOD correlated positively with alkalinity (p -value < 0.0001 and $r = 0.74$), $\text{NH}_4\text{-N}$ (p -value < 0.0001 and $r = 0.74$), $\text{NO}_3\text{-N}$ (p -value < 0.001 and $r = 0.70$), and negatively with DO (p -value < 0.001 and $r = 0.68$) (Spearman’s rank correlation). DO correlates negatively with the absolute levels of *yccT*, relative levels of *sul1*, COD, BOD, $\text{NH}_4\text{-N}$, $\text{NO}_3\text{-N}$, $\text{PO}_4\text{-P}$, and chlorophyll-*a* (p -value < 0.05 ; $r = 0.50$) (Spearman’s rank correlation) (Figure 2, Table S5). Alkalinity correlated

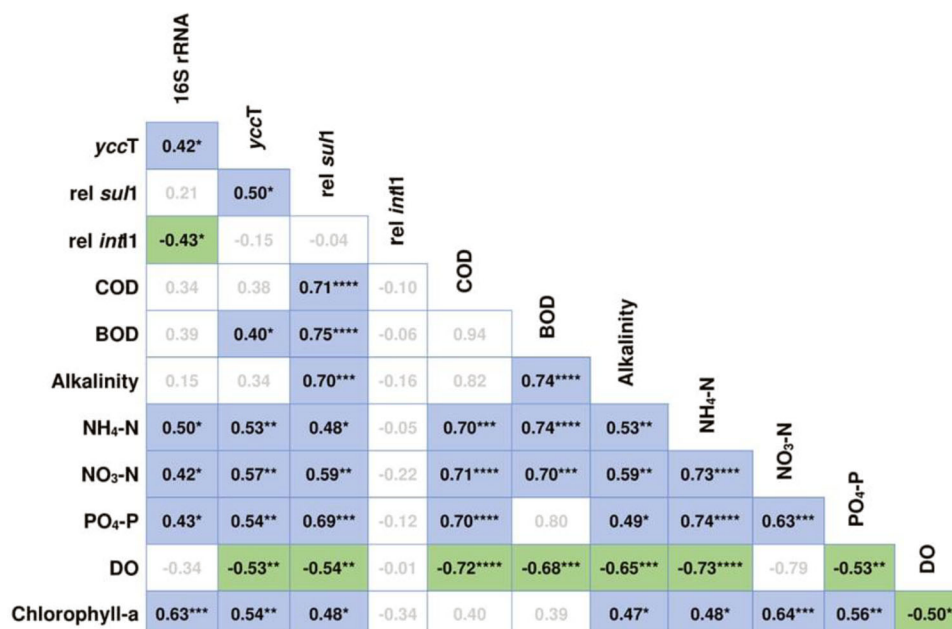


FIGURE 2 Correlation matrix: Spearman's rank correlation coefficients (ρ) among the levels of targeted genes and physicochemical parameters for all ponds combined (* p -value < 0.05, ** p -value < 0.01, *** p -value < 0.001, and **** p -value < 0.0001). In this figure, blue and green areas represent the positive and negative correlation between the levels of targeted genes and physicochemical parameters, respectively.

positively with NH₄-N (p -value < 0.01; r = 0.53), NO₃-N (p -value < 0.01; r = 0.59), PO₄-P (p -value < 0.05; r = 0.49), and chlorophyll-*a* (p -value < 0.05; r = 0.47) and negatively with DO (p -value < 0.001; r = 0.65) (Spearman's rank correlation) (Figure 2, Table S5). The levels of NH₄-N correlated positively with NO₃-N (p -value < 0.0001 and r = 0.73), PO₄-P (p -value < 0.0001 and r = 0.74), and chlorophyll-*a* (p -value < 0.05 and r = 0.48) and negatively with DO (p -value < 0.0001 and r = 0.73) (Spearman's rank correlation) (Figure 2, Table S5). The NO₃-N levels correlated positively and significantly with PO₄-P (p -value < 0.001 and r = 0.63) and chlorophyll-*a* (p -value < 0.001 and r = 0.64) (Spearman's rank correlation). The PO₄-P levels correlated positively and significantly with chlorophyll-*a* (p -value < 0.01 and r = 0.56) and negatively with DO (p -value < 0.01 and r = 0.53) (Spearman's rank correlation, Figure 2, Table S5).

3.2 | Levels of the DNA markers of total bacterial count and fecal indicator

3.2.1 | 16S rRNA gene

The 16S rRNA gene copy levels were used to track trends in the total bacterial count in the samples. In the water samples from the WRPs, the levels of 16S rRNA gene copies were approximately 1–2 orders of magnitude higher relative to the ponds (p -value = 0.007, Mann–Whitney U test; Figure 3A). Overall, the levels of 16S rRNA gene copies

in the water correlated positively and significantly with the levels of *yccT* (p -value < 0.05 and r = 0.42), NH₄-N (p -value < 0.05 and r = 0.50), NO₃-N (p -value < 0.05 and r = 0.42), and chlorophyll-*a* (p -value < 0.001 and r = 0.63) (Spearman's rank correlation) (Figure 2, Table S5) and correlated negatively with relative levels of *int1* (p -value < 0.01 and r = 0.43) (Spearman's rank correlation) (Figure 2, Table S5). In both the sediment and soil samples, the levels of 16S rRNA gene copies were not statistically different across the sites (p -value > 0.05, Mann–Whitney U test; Figure 3B,C).

3.2.2 | Fecal indicator (*E. coli*), *yccT*

The absolute levels of the DNA marker of *E. coli*, *yccT*, were nearly an order of magnitude higher in the WRPs relative to the water in the ponds (p -value = 0.01, Mann–Whitney U test; Figure 3D). The WRP that received wastewater from a dairy processing unit (site G) had *yccT* levels comparable to other WRPs (p -value = 0.43, Mann–Whitney U test; Figure 3D). The *yccT* levels in the water samples from both the WRPs and the ponds trended as P–I < P–P < WRPs (Figure 3D). The relative levels of *yccT* ranged from –3 to 0 on the log₁₀ scale (0.1%–100% of total bacterial count) (Figure S3), indicating very high levels of fecal contamination in all water bodies.

In the sediment and the soil samples, the absolute levels of *yccT* were comparable between the WRPs and the ponds (p -value > 0.05, Mann–Whitney U test; Figure 3E,F). The ponds had a higher absolute level of *yccT* in the adjoining

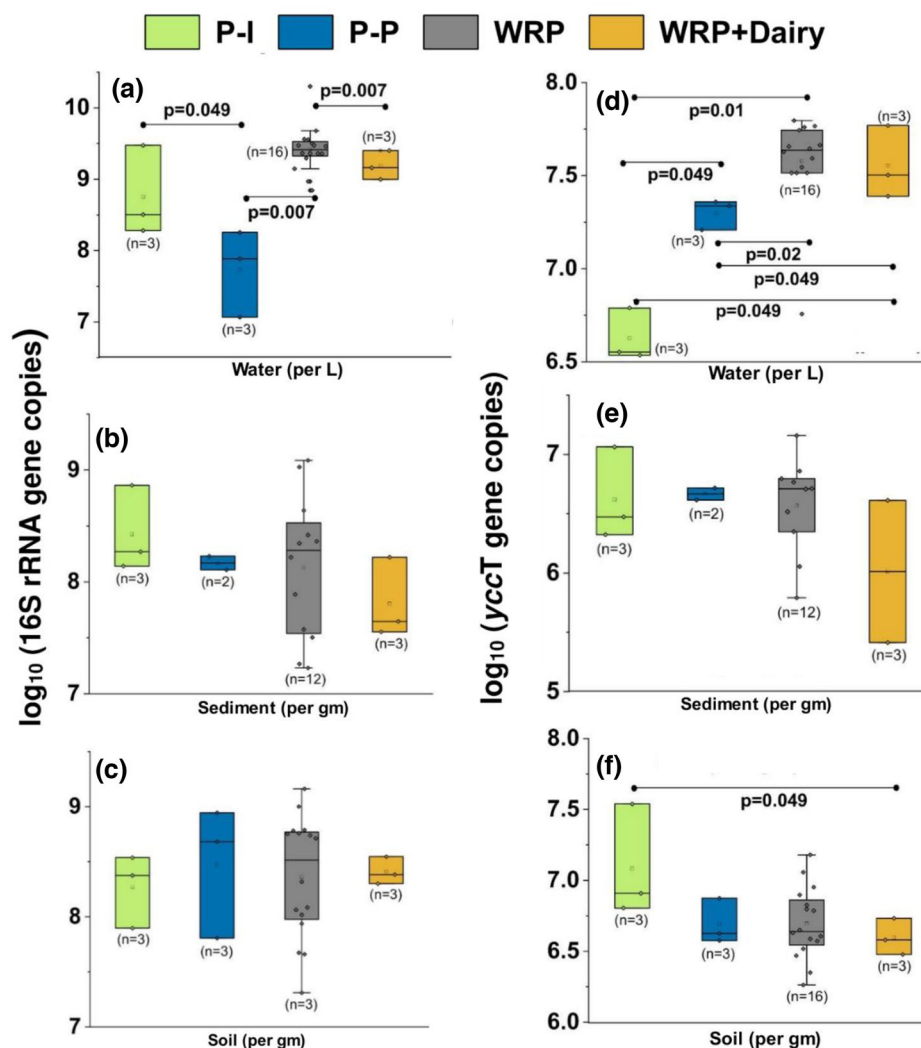


FIGURE 3 Total bacterial count and *Escherichia coli*: The y-axis represents the absolute levels of (A–C) 16S rRNA gene copies and (D–F) *yccT* gene copies per liter of sampled water, and per gram of sediment and soil on a log₁₀ scale. The x-axis represents the control pond, whose water was used for irrigation (P–I); the control pond, which was used for pisciculture (P–P), wastewater-receiving ponds (WRPs), and WRP at site G, which also received wastewater from a dairy-processing unit (WRP + dairy). The statistically significant difference, as determined by a *p*-value of <0.05, is indicated directly above the horizontal straight line between the sample sets that were compared.

soil than the WRPs (Figure 3F) but had lower absolute levels of *yccT* than the WRPs in the sediment sample (Figure 3E). The relative levels of *yccT* in the water were nearly an order of magnitude higher in the P–P samples relative to the WRPs (*p*-value = 0.0073, Mann–Whitney *U* test; Figure S3) and 1–2 orders of magnitude higher relative to the P–I water samples (*p*-value = 0.049, Mann–Whitney *U* test; Figure S3).

The *yccT* (marker of *E. coli*) correlated positively and significantly with the relative levels of *sul1* (*p*-value < 0.05; *r* = 0.50), BOD (*p*-value < 0.05; *r* = 0.40), NH₄–N (*p*-value < 0.01; *r* = 0.53), NO₃–N (*p*-value < 0.01; *r* = 0.57), PO₄–P (*p*-value < 0.01; *r* = 0.54), and chlorophyll-*a* (*p*-value < 0.01; *r* = 0.54) and negatively with DO (*p*-value < 0.01, *r* = 0.53; Spearman's rank correlation) (Figure 2, Table S5).

3.3 | Levels of anthropogenic antibiotic resistance gene markers

3.3.1 | Levels of *sul1*

The absolute levels of *sul1* in the water in the WRPs (sites A–F) ($10^{7.5}$ – $10^{8.5}$ gene copies/L) had higher levels than in the control ponds (sites H and I) (10^6 – $10^{6.5}$ gene copies/L) and the WRP at site G ($10^{6.6}$ gene copies/L) (Mann–Whitney *U* test; Figure 4A). In the sediment samples, the absolute levels of *sul1* were higher in WRPs and P–P relative to P–I, (*p*-value < 0.05, Mann–Whitney *U* test; Figure 4B). The absolute levels of *sul1* in the adjoining soil trended as WRPs ($10^{5.6}$ – $10^{10.1}$ gene copies/g) > ponds (P–I and P–P) ($10^{5.6}$ – 10^7 gene copies/g) (*p*-value = 0.02, Mann–Whitney *U* test; Figure 4C).

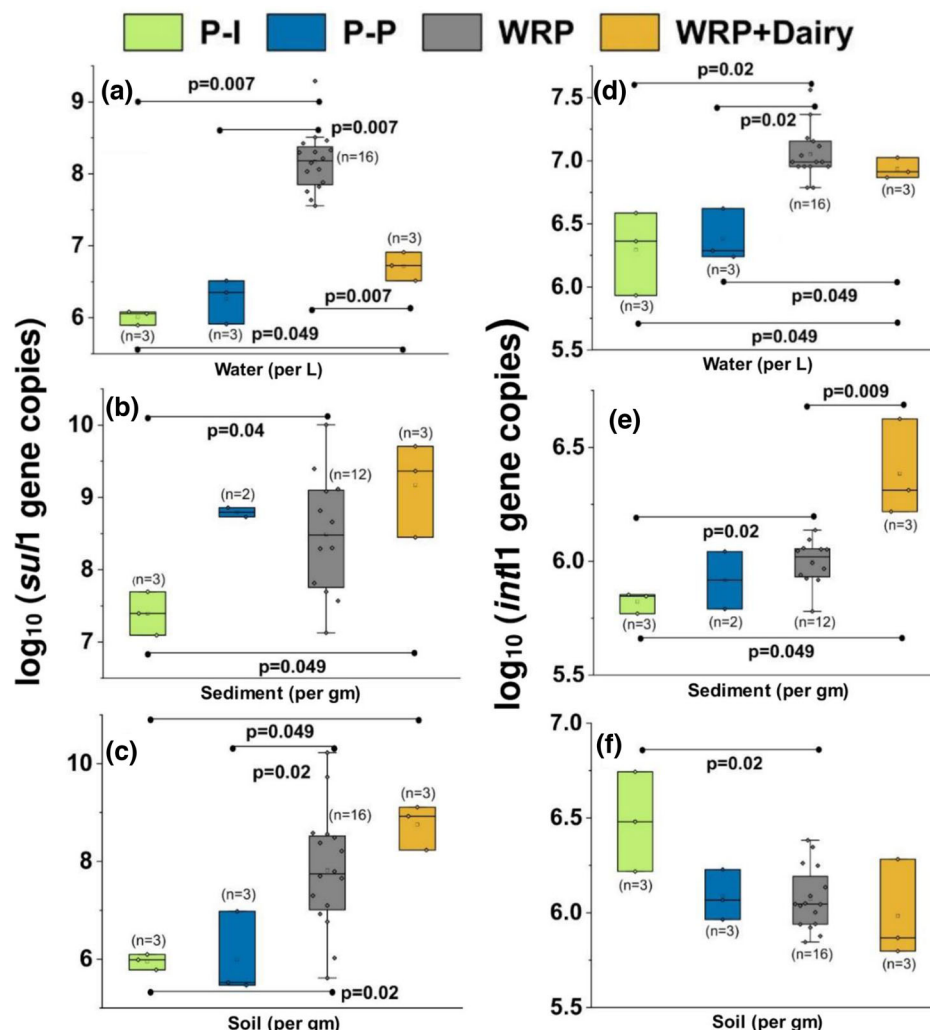


FIGURE 4 Levels of anthropogenic antibiotic resistance gene markers: The y-axis represents the absolute levels of (A–C) *sul1* and (D–F) *int11* gene copies per liter of sampled water, and per gram of sediment and soil on a log₁₀ scale. The x-axis represents the control pond, whose water was used for irrigation (P-I); the control pond, which was used for pisciculture (P-P); wastewater receiving ponds (WRPs); and WRP at site G, which also received wastewater from a dairy-processing unit (WRP+dairy). The statistically significant difference, as determined by a *p*-value of <0.05, indicated directly above the horizontal straight line between the sample sets that were compared.

The relative levels of *sul1* in the water of WRPs and P-P (1%–10% of total bacterial count) were about an order of magnitude higher relative to the levels in P-I and the WRP at site G (0.03%–0.32%–100% of total bacterial count) (*p*-value = 0.007, Mann–Whitney *U* test; Figure S4A, Section SI.2.1). Similarly, the relative levels of *sul1* in the sediment were 1–2 orders of magnitude higher in the WRPs and P-P relative to the P-I samples (*p*-value = 0.03, Mann–Whitney *U* test; Figure S4B, Section SI.2.1). The relative levels of *sul1* in the local soil were 2–3 orders of magnitude higher in the WRPs samples relative to the control ponds (P-I: *p*-value = 0.033, Mann–Whitney *U* test, Figure S4C; and P-P: *p*-value = 0.044, Mann–Whitney *U* test; Figure S4C, Section SI.2.1). The relative levels of *sul1* correlated positively with COD (*p*-value < 0.0001; *r* = 0.71), BOD (*p*-value < 0.0001; *r* = 0.75), alkalinity (*p*-value < 0.001;

r = 0.70), NH₄-N (*p*-value < 0.05; *r* = 0.48), NO₃-N (*p*-value < 0.01; *r* = 0.59), PO₄-P (*p*-value < 0.001; *r* = 0.69), and chlorophyll-*a* (*p*-value < 0.05; *r* = 0.48), and negatively with DO (*p*-value < 0.01; *r* = 0.54) (Figure 2, Table S5).

3.3.2 | Levels of *int11*

The absolute levels of *int11* in the water were higher in samples from the WRPs ($10^{6.75}$ – $10^{7.3}$ gene copies/L) relative to the ponds ($10^{5.9}$ – $10^{6.6}$ gene copies/L) (*p*-value = 0.02, Mann–Whitney *U* test; Figure 4D). In the sediment, the absolute levels of *int11* were highest in the WRP at site G, which also received wastewater from a dairy processing unit ($10^{6.2}$ – $10^{6.5}$ gene copies/g) and were lowest in the control pond at site H (P-I) ($10^{5.75}$ – $10^{5.8}$ gene copies/g) (Figure 4E). The absolute

levels of *intI1* in the soil adjacent to the water bodies were not statistically different among the water bodies. However, the relative levels of *intI1* in the water were nearly an order of magnitude higher in the P–P samples (1%–20% of total bacterial count) relative to the WRPs and the ponds (0.08%–2.5% of total bacterial count) (p -value = 0.01, Mann–Whitney U test; Figure 4D). The mean of the relative levels of *intI1* in the sediment trended as P–I < P–P < WRPs (except for site G) < site G (Figure S4E, Section SI.2.2). However, only the relative levels of *intI1* in the sediment of P–I (–2.3 to –3.1 on log10 scale) was statistically significantly lower relative to that of site G (–0.8 to –2 on log10 scale) (p -value = 0.049; Mann–Whitney U test; Figure S4E). The relative levels of *intI1* in the soil adjacent to the ponds were higher than in the case of the WRP at site G but were not statistically different from other water bodies (WRPs and P–P) (p -value = 0.01, Mann–Whitney U test; Figure S4F, Section SI.2.2). Interestingly, there was no significant correlation between the relative levels of *intI1* and water quality parameters (Figure 2, Table S5).

4 | DISCUSSION

4.1 | The water quality in WRPs was worse than in domestic sewage

The water in the WRPs was worse or comparable in terms of water quality parameters to the sewage typically received by the sewage treatment plants in the region (Table S4). The extent of fecal contamination and subsequent deterioration of water quality observed in these seven WRPs is the worst ever reported for any surface water body (Dehghani et al., 2018; Xu et al., 2022). The impact of residing in proximity of these extremely contaminated surface water bodies on public health burden is not yet known; however elsewhere, living close to other fecally contaminated surface water bodies has been linked with a higher prevalence of waterborne diseases such as typhoid, diarrhea (Null et al., 2018), hepatitis, skin diseases (Khan et al., 2018), malnutrition and dysentery (Thys et al., 2015), and respiratory infections (Mataloni et al., 2016).

In addition to possibly posing a public health threat, the extreme contamination and eutrophication in these surface water bodies, which have traditionally provided water resilience (Wang et al., 2021) and opportunities for pisciculture (Hamza et al., 2020) to rural communities, could drive their destruction (Wurtsbaugh et al., 2019). The higher levels of chlorophyll-*a* in WRPs correlated with fecal contamination indicating that eutrophication and accelerated disappearance of the water body were driven by fecal contamination (Abis & Mara, 2005; Ji et al., 2018). The water quality parameters and total bacterial count of the water in the WRPs were comparable to that of raw sewage, which indicates that the freshwater

sources of the WRPs are drying up, and domestic wastewater is the primary source of water. With increasing water scarcity and a rapidly changing climate (Katyaini & Barua, 2017; Zhao et al., 2019), the disappearance of ponds due to extreme fecal contamination could threaten water resilience (Dehghani et al., 2018; Katyaini & Barua, 2017; Zhao et al., 2019).

As the DO of the WRPs was <2 mg/L, setting-in of anaerobic conditions (Dien et al., 2018), obliteration of aquatic biodiversity (Middleton & Reeder, 2003; Takekawa et al., 2015), and anaerobic putrefaction are likely. The extreme contamination and anoxic conditions observed in the WRPs herein suggest that we were already past the initial contamination and massive fish kill events in these WRPs. As only the ponds that did not receive sewage were viable for use in irrigation and pisciculture, it could be surmised that extreme fecal contamination has shut down avenues for pisciculture and irrigation in the region.

4.2 | WRPs fail to treat domestic wastewater

Unlike wastewater stabilization ponds, the WRPs are not engineered for wastewater treatment and only serve as a convenient, albeit ineffective, environment for disposing of wastewater. Although adequate sewage treatment cannot be expected from WRPs, tracking the persistence of fecal microbes and the water quality in the WRPs is essential to protect public health. The levels of 16S rRNA gene copies and the water quality parameters, such as BOD, COD, N, and P in the WRPs, were comparable to that of medium-to high-strength raw sewage reported elsewhere (Dehghani et al., 2018; Jiang et al., 2021; Ospanov et al., 2020). It is evident from the data gathered herein that the seven WRPs investigated herein were mere dumping sites for the community's wastewater. In this context, the WRPs potentially present a severe potential threat to public health (Mthembu et al., 2010).

The WRPs pool the community's wastewater and are typically much closer to residences than open defecation sites have been in the past. Such proximity of WRPs to the homes (2–40 m) increases the possibility of a considerable public health burden (Gorham et al., 2017; Khan et al., 2018). It is notable that the WRPs are contaminated environments that are accessible by both stray and community animals, and humans and thus present a 'One Health' challenge. As the WRPs were nutrient-rich and had very high levels of fecal indicator (as indicated by the high relative levels of *ycdT*: –3 to 0 on the log10 scale) (0.1%–100% of total bacterial count), it is reasonable to suspect that the fecal microbes constituted a large portion of the microbial count in the WRPs. As there is evidence of no wastewater treatment and persistence of fecal indicators in these WRPs, the fecally–orally transmitted pathogens may survive long enough in WRPs for vector-borne

transmission of diseases to the community and render the decision to build bathrooms and latrines without providing adequate infrastructure for wastewater treatment questionable from a public health perspective. On the other hand, the pond at the site I, which was used both for pisciculture and bathing by local children, had fecal deposits in its vicinity, suggesting open defecation (Jangra et al., 2016). Open defecation, while having a relatively lesser impact on the surface water quality relative to direct sewage disposal in WRPs, as noted herein, still adversely impacts public health. This implies that neither overlooking open defecation nor building toilets sans wastewater treatment facilities is meaningful in the context of improving public and environmental health.

It is worth noting that in contrast to all the WRPs, the two ponds were under direct private care and did not receive either the sewage or municipal solid waste and consequently had better water quality fit for irrigation and pisciculture (Baghel et al., 2005; Xu et al., 2022). The ownership and responsibility for the upkeep of the surface water body affect the water quality and utility of water bodies.

4.3 | Anthropogenic antibiotic resistance gene markers were elevated in WRPs, and the relative levels of *sul1* were linked with fecal contamination

Both fecal and environmental contamination with co-selective agents, such as antibiotics and certain heavy metals (Baker-Austin et al., 2006; Karkman et al., 2019; Li et al., 2017; Seiler & Berendonk, 2012), enrich the antibiotic-resistance in the environmental microbial communities. The WRPs pool not only the sewage with its high bacterial count and nutrient levels but also co-selective agents, such as unmetabolized and partially metabolized antibiotics. Consequently, the WRPs are potential breeding and selection grounds for antibiotic resistance (Binh et al., 2018; Neudorf et al., 2017; Wang et al., 2016). The trends in the levels of these DNA markers are a reasonably good estimate of actual trends in the levels of antibiotic resistance in the environment (Peterson & Kaur, 2018; Sultan et al., 2018). Both *sul1* and *int11* have recently been established as an indicator of a human-impact-induced increase in antibiotic resistance in environmental microbial communities (Colomer-Lluch et al., 2014; Le et al., 2016). Although the levels of *sul1* indicate overall trends in antibiotic resistance potential, class 1 integron-integrase gene cassette, *int11*, is also (a) associated with elevated levels of resistance to heavy metals (Bhaskar & Singh, 2021; Lu et al., 2015); (b) regarded to be indicative of the primary mode of integration of the horizontally transferred antibiotic-resistant genes (Chen et al., 2015; Stecher et al., 2013; Willems et al., 2011); and (c) considered to provide a competitive edge to bacterial populations in stressful environments (Gillings et al., 2015).

Overall, the levels of *sul1* correlated positively with physicochemical parameters that increase when untreated wastewater is added to the water bodies (BOD, COD, alkalinity, N, P, and *ycrT*) and negatively with DO, which decreases with increased sewage loading. This indicates a link between elevated *sul1* levels, fecal contamination, and deterioration of water quality. The high nutrient levels and the co-selective agents typically present in untreated wastewater could synergistically impact the antibiotic resistance potential in the WRPs (Subirats et al., 2018). Interestingly, in the pond used for pisciculture, the high relative abundance of the *sul1* and *int11* indicates that the practice of pisciculture in the region could also be an important driver of antibiotic resistance potential. It is possible that co-selective agents of antibiotic resistance were used in the practice of pisciculture, perhaps from the use of poultry leftovers as feed, but relevant information was unavailable. As this pond was also used for swimming and washing by the residents, exposure to antibiotic-resistant bacteria via dermal contact and accidental ingestion is possible.

As the hydraulic retention time of the WRPs is infinite (the water is never drained out, only evaporated due to solar irradiation, or possibly absorbed into the ground), the WRPs could be a long-term reservoir for antibiotic-resistance potential and pathogens, providing ample opportunity for any horizontal transfer of antibiotic resistance to pathogens and vector-borne transmission of diseases.

5 | CONCLUSION

When an adequate wastewater treatment system does not accompany toilet construction, the wastewater is discharged into low-lying areas and local ponds. The water quality in seven WRPs was comparable to high-strength raw sewage in terms of BOD, COD, N, P, and fecal indicator levels. The relative levels of anthropogenic antibiotic resistance gene markers were elevated in WRPs relative to ponds that did not receive domestic wastewater. The fecal indicator correlated with relative levels of *sul1*, BOD, COD, chlorophyll-*a*, and nutrient levels, implying that fecal contamination is a significant driver of water quality deterioration and overall antibiotic resistance potential in WRPs. The results highlight the futility of constructing toilets without adequate wastewater treatment or management systems to prevent environmental contamination with fecal matter and protect public health.

AUTHOR CONTRIBUTIONS

Neelam Gunjyal: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Gargi Singh:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources,

Supervision, Writing – review & editing. **Chandra Shekhar Prasad Ojha**: Supervision.

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CONFLICT OF INTEREST STATEMENT

All the authors declare that they have no recognized conflict of interests or personal relationships that could have appeared to influence the work reported in this paper.

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SUPPORTING INFORMATION

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